

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester MCA (CBCS) Degree Examinations
 December 2024

22MCA302 – ADVANCED WEB TECHNOLOGIES

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Provide a brief overview on the usage of the Jumbotron in Bootstrap.	5	L*2	1	1
b)	What are semantic elements in HTML5? Explain any five of them.	7	L2	1	1
c)	Explain different levels of applying styles to an HTML document using CSS. Also explain various Selector forms supported by CSS.	8	L2	1	1
2. a)	Explain the Box Model properties in CSS with suitable examples.	5	L2	1	1
b)	What is the Bootstrap Grid System? Explain its operation.	7	L2	1	1
c)	Explain following with respect to HTML5.				
i.	Computer code elements				
ii.	New form elements	8	L2	1	3
Unit – II					
3. a)	List the differences between				
i.	GET and POST methods				
ii.	Sessions and Cookies	10	L1	2	1
b)	Write a PHP program to demonstrate following operations with arrays				
i.	Creating arrays (indexed and associative)				
ii.	Accessing array elements				
iii.	Modifying arrays				
iv.	Looping through arrays				
v.	Common array functions	10	L3	2	1
4. a)	Provide steps involved in Session tracking in PHP. Also provide an example that demonstrates session tracking for a user login system.	10	L2	2	1
b)	Write a PHP program to connect to a MySQL database consisting of user's details id, email and name and carries out basic CRUD operations.	10	L3	2	3
Unit – III					
5. a)	Explain how values are converted between different types in JavaScript.	6	L2	3	1
b)	Describe the concept of anonymous functions and arrow functions in JavaScript, providing appropriate examples for each.	7	L3	3	3
c)	What are Document Type Definitions (DTDs) in XML? Provide an example to demonstrate the declaration of elements and attributes in a DTD.	7	L3	3	1
6. a)	Compare search() and match() methods provided by String object in JavaScript for pattern matching.	6	L3	3	1
b)	Show the use of methods provided by Array object in JavaScript.	6	L2	3	1
c)	What is an XML schema? Show how it is implemented.	8	L3	3	3
Unit – IV					
7. a)	Describe the DOM2 event model and provide relevant examples to illustrate its functionality.	10	L2	4	1
b)	Elaborate the CSS property position. Provide a JavaScript example to show how position property and directional properties can be used to move an element in up, down, left, right directions.	10	L4	4	4

P.T.O.

22MCA302

SEE -December 2024 – January 2025

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|-------|--|----|----|---|---|
| 8. a) | With suitable examples, describe various techniques available in JavaScript for accessing elements. | 10 | L3 | 4 | 1 |
| b) | Create an HTML document to illustrate the concept of dynamic stacking of images-Three images are placed on the document so that they overlap; when a user clicks on any image, it must come to the top position. | 10 | L3 | 4 | 3 |

Unit – V

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|--------|--|----|----|---|---|
| 9. a) | Outline the various selectors available in the jQuery library. | 5 | L2 | 5 | 1 |
| b) | What is callback function in jQuery? Explain with an example. | 5 | L2 | 5 | 3 |
| c) | Discuss the purpose and functionality of modules and controllers in AngularJS. | 10 | L4 | 5 | 1 |
| 10. a) | Describe various methods provided by jQuery library to | | | | |
| | i. Add effects on webpages | | | | |
| | ii. Add and remove contents in jQuery. | 10 | L3 | 5 | 1 |
| b) | What are filters in AngularJS? Explain any five built in filters supported by AngularJS. Also show how a custom filter can be created to reverse a string. | 10 | L4 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
